

FLORA, IL, USA

WMO: 744658

Lat: 38.665N

Lon: 88.453W

Elev: 144

StdP: 99.61

Time Zone: -6.00 (NAC)

Period: 02-19

WBAN: 53889

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-14.0	-11.2	-19.1	0.7	-13.1	-16.5	0.9	-10.7	11.9	5.2	10.8	5.2	2.7	300	0.463

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.8	34.9	24.6	33.1	24.2	32.1	23.8	26.8	31.3	25.8	30.5	25.0	29.7	3.9	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.7	21.3	29.7	24.6	19.9	28.7	23.6	18.7	27.6	85.1	31.3	80.5	30.8	76.7	29.5	31.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.2	8.6	7.7	-18.0	36.2	3.8	1.9	-20.8	37.6	-23.0	38.7	-25.1	39.8	-27.8	41.2		
(5)			-18.4	27.6	3.5	1.3	-20.9	28.6	-23.0	29.4	-25.0	30.1	-27.5	31.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	13.3	-0.3	1.2	7.6	13.5	18.9	23.8	25.1	24.2	20.7	14.2	7.5	2.4		
	DBStd	10.23	6.38	6.33	6.02	5.02	4.61	3.14	3.04	2.95	3.76	5.05	5.26	5.37		
	HDD10.0	1086	323	254	119	23	1	0	0	0	15	108	243			
	HDD18.3	2592	577	480	334	155	52	2	1	2	18	150	326	495		
	CDD10.0	2288	4	7	46	129	277	415	467	439	320	144	33	6		
	CDD18.3	752	0	0	3	11	70	167	209	182	88	21	1	0		
	CDH23.3	7382	0	0	22	119	623	1618	2138	1745	889	221	8	0		
	CDH26.7	2917	0	0	2	14	193	655	933	721	338	61	1	0		
(14) Wind	WSAvg	3.1	4.1	3.8	3.9	3.9	3.1	2.7	2.2	2.0	2.1	2.9	3.4	3.7		
(15) Precipitation	PrecAvg	1131	89	61	90	107	133	107	99	89	79	86	106	75		
	PrecMax	1507	194	122	220	236	287	213	201	180	231	194	228	184		
	PrecMin	908	9	11	22	32	36	29	45	34	6	21	4	39		
	PrecStd	163	48	31	44	58	71	52	45	46	53	50	52	37		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.9	20.8	26.3	28.0	32.6	35.3	38.0	36.5	35.0	31.3	24.0	18.7		
		MCWB	15.0	15.0	17.2	18.7	22.1	24.1	23.7	24.8	21.7	21.5	16.4	15.5		
	2%	DB	15.1	17.3	22.5	26.5	30.4	33.6	34.8	34.1	32.6	28.0	21.2	15.6		
		MCWB	12.4	13.8	16.1	17.7	21.7	23.9	25.7	24.3	22.8	19.4	15.9	13.5		
	5%	DB	12.5	14.0	20.0	24.6	28.9	32.3	32.9	32.4	30.7	25.9	18.6	12.7		
		MCWB	10.4	10.6	14.5	17.0	20.9	23.4	25.1	24.1	22.3	18.2	14.4	10.2		
	10%	DB	9.1	11.0	17.5	22.4	27.3	30.5	31.8	30.8	28.0	23.0	16.3	10.6		
		MCWB	7.0	8.2	13.5	16.1	20.1	22.5	24.4	23.7	20.4	17.4	12.6	8.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.6	16.9	18.5	20.9	24.3	27.3	28.8	28.7	25.4	23.2	19.4	16.6		
		MCDB	17.5	19.7	23.5	25.9	29.6	31.8	33.2	34.4	30.7	28.1	22.6	18.1		
	2%	WB	13.4	14.4	17.0	19.3	23.0	25.6	27.2	26.5	24.4	21.9	17.3	14.0		
		MCDB	14.7	16.7	21.6	24.3	28.2	31.2	31.7	30.8	29.3	26.5	19.3	15.4		
	5%	WB	10.7	11.2	15.6	18.1	22.0	24.6	26.2	25.4	23.3	20.2	15.3	11.1		
		MCDB	11.9	13.4	19.0	22.7	26.9	29.8	30.7	30.1	27.8	23.8	18.4	12.4		
	10%	WB	6.9	7.9	13.9	16.9	21.0	23.7	25.3	24.5	22.2	18.4	12.7	8.6		
		MCDB	9.0	10.4	17.1	21.1	25.7	28.4	29.7	28.9	26.7	21.8	15.6	9.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.4	8.8	10.1	11.4	10.7	11.0	10.8	11.5	12.7	12.1	10.2	7.9		
		MCDBR	11.9	13.4	13.4	14.2	12.7	12.8	12.1	13.4	15.1	15.0	14.0	11.0		
	5% WB	MCWBR	10.0	10.4	8.3	7.8	5.5	4.9	4.7	5.3	5.8	7.2	9.2	9.0		
		MCWBR	10.9	12.6	11.9	12.1	11.1	10.7	10.5	11.1	12.0	11.9	11.8	9.9		
(40) Clear-Sky Solar Irradiance	taub		0.309	0.323	0.360	0.402	0.438	0.439	0.464	0.465	0.412	0.360	0.320	0.307		
		taud	2.453	2.400	2.364	2.247	2.204	2.224	2.157	2.141	2.281	2.425	2.484	2.471		
	Ebn at Noon		868	902	898	875	847	844	818	807	831	848	849	843		
		Edh at Noon	81	98	112	134	143	141	149	148	121	95	79	75		
(44) All-Sky Solar Radiation	RadAvg		1.94	2.73	3.71	4.89	5.51	6.39	6.20	5.62	4.69	3.38	2.27	1.65		
	RadStd		0.19	0.29	0.34	0.28	0.48	0.41	0.39	0.32	0.33	0.35	0.25	0.16		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (42 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+78

Nomenclature: See separate page